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Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology
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Avadi, Chennai



Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

END SEMESTER PROJECT REVIEW

Date: 06-05-2026

SMART CAMPUS EVENT MANAGEMENT SYSTEM

SDG: SDG No - SDG4 – Quality Education

Presented By:

A.SUSMITHA – VTU25984 – 23UECS0021

Courses Handling Faculty:

Dr. HAFIZUR RAHMAN

Assistant Professor



INTRODUCTION

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Smart Campus Event Management System is a web-based application for managing college events

Provides a centralized platform for students and administrators

Students can:

- Register online
- Track their participation

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Admins can:

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- Add, edit, and delete events
- Manage event details
- View upcoming events
- View registration statistics

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Reduces manual work and improves event coordination

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Built using:

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- Spring Boot, Spring MVC
- Spring Data JPA
- Thymeleaf, HTML, CSS

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Ensures secure, scalable, and user-friendly system

Supports United Nations SDG 4 – Quality Education



PROBLEM STATEMENT

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- In many educational institutions, campus events such as workshops, seminars, and technical programs are managed using manual methods or fragmented digital platforms.
- Students often face challenges in registering for events and tracking their participation, leading to reduced engagement.
- Administrators encounter issues in efficiently managing event details, handling registrations, and maintaining records.
- The absence of a centralized platform results in:
 - Ineffective communication
 - Increased chances of errors and data inconsistency
 - Difficulty in monitoring and analyzing event performance
- Additionally, there is no streamlined way to generate insights such as registration statistics or event success metrics.
- This lack of a unified system creates difficulties for students in accessing accurate, timely, and organized event information.
- Therefore, there is a need for a centralized, automated, and user-friendly system that simplifies event management and enhances student participation.



OBJECTIVE(S)

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- Enable students to view and register for events easily
- Provide admins with tools to create, update, and delete events
- Improve communication and participation in campus activities
- Ensure a secure and user-friendly platform
- Support United Nations SDG 4 – Quality Education
- Develop a centralized system to manage campus events



PROPOSED SYSTEM

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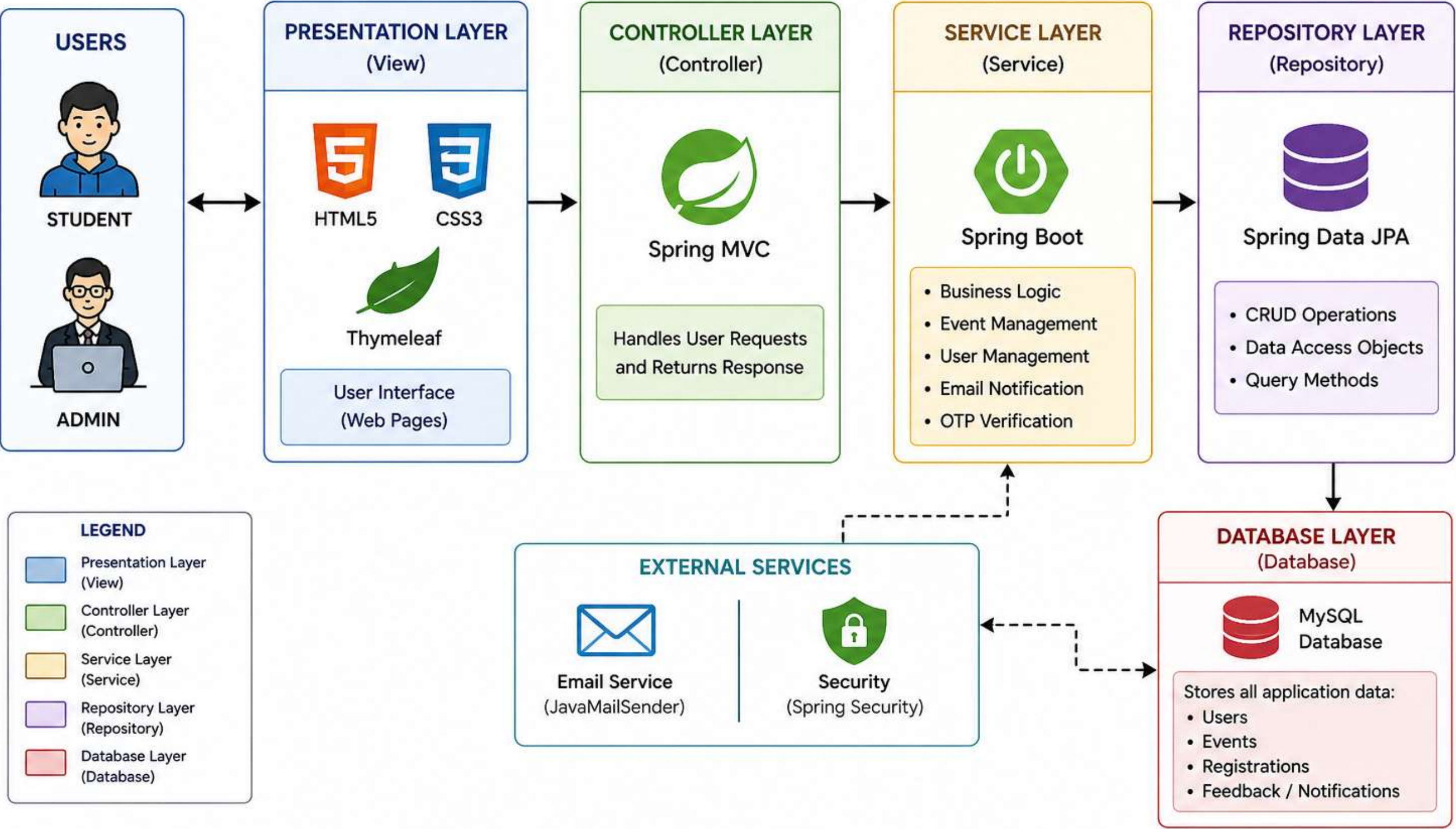
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- Provides a real-time event dashboard showing upcoming and trending events
- Introduces a role-based system with separate access for students and administrators
- Implements automated registration handling, reducing delays and manual errors
- Supports event capacity management (limit number of participants)
- Enables feedback collection after events to improve future planning
- Includes notification features (alerts/reminders for registered events)
- Offers data analytics such as participation trends and popular event categories
- Ensures scalability, allowing the system to handle increasing users and events
- Designed with a responsive interface, making it accessible on mobile and desktop

PROPOSED SYSTEM ARCHITECTURE



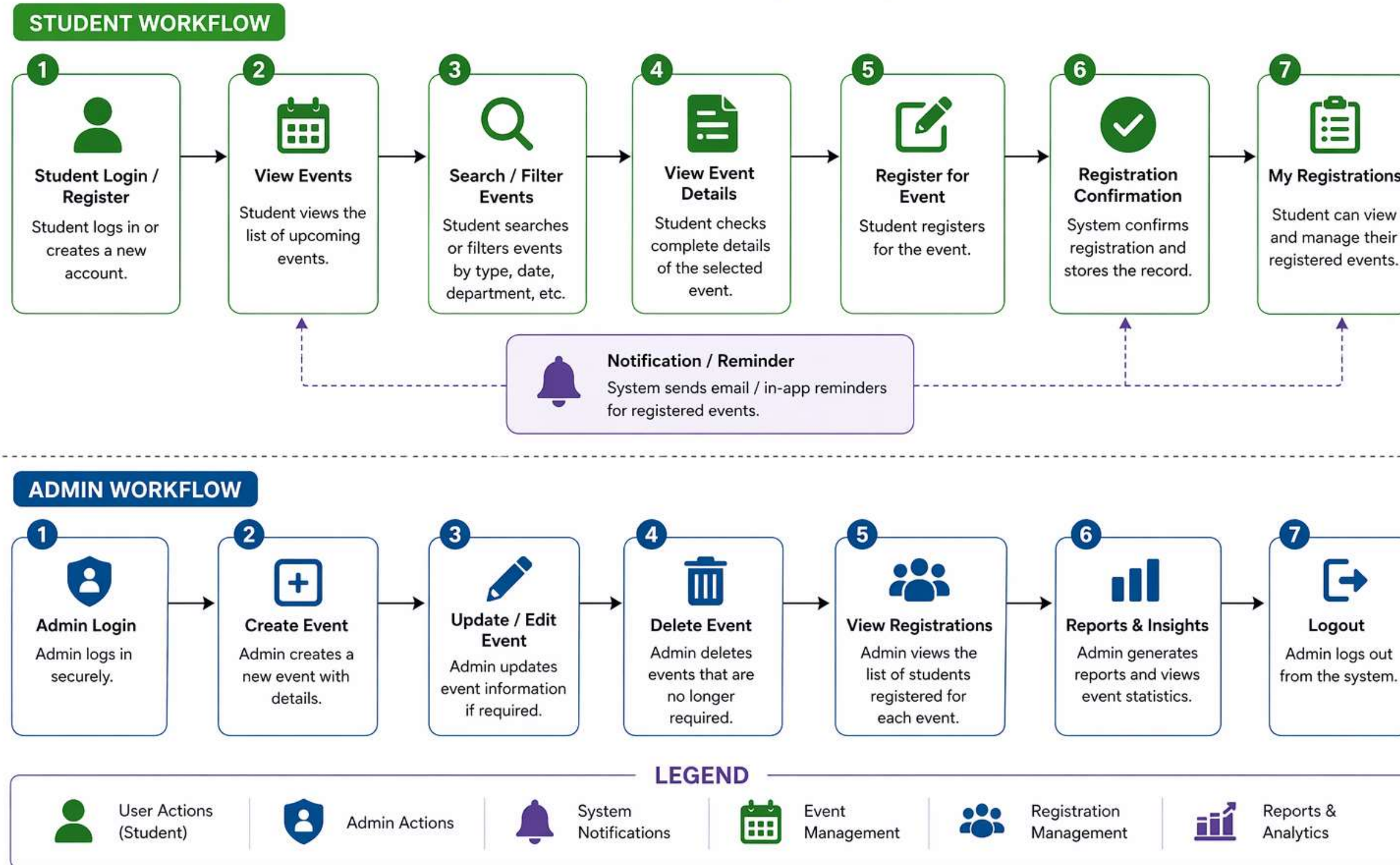
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WORKFLOW

Smart Campus Event Management System



TECHNOLOGIES USED



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- Frontend:

HTML5, CSS3, Thymeleaf, Bootstrap

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- Database:

MySQL

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- Backend:

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Java, Spring Boot, Spring MVC

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- ORM Framework:

Spring Data JPA (Hibernate)

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- Security:

Spring Security (Basic Authentication)

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- Build Tool:

Maven

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- Server:

Embedded Tomcat Server

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- Tools/IDE:

Visual Studio Code



MODULE DESCRIPTION

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❖ **Module 1: Student Module**

- Browse and search events
- Register for events
- View registered events
- Receive notifications/reminders

❖ **Module 2: Admin Module**

- Secure login and authentication
- Add, update, delete events (CRUD)
- Manage event details and registrations
- View reports and analytics



MODULE DESCRIPTION

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❖ **Module 3: Event Management Module**

- Create and manage event data
- Search and filter events (date, type, department)
- Handle event capacity and details

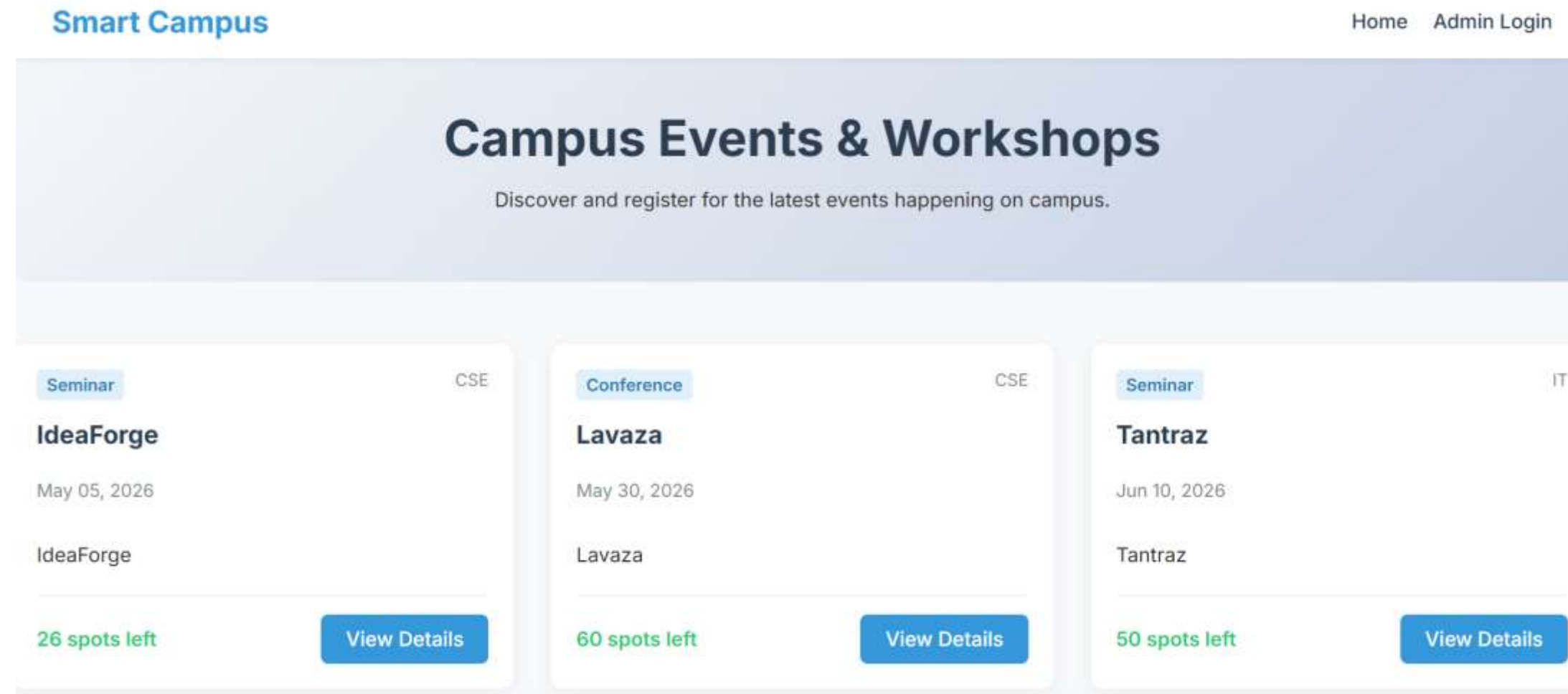
❖ **Module 4: Database & Security Module**

- Store event, student, and registration data
- Ensure data consistency and integrity
- Provide authentication and authorization
- Secure system access



IMPLEMENTATION SCREENSHOTS

- Home Page



- The Home page displays the list of upcoming events available in the Smart Campus Event Management System.
- It allows users to view event details such as event name, date, location, and status, and provides options to manage events.

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8(A)

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IMPLEMENTATION SCREENSHOTS

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- Admin Login Page

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8(B)

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Admin Login

Username

Password

Sign In



IMPLEMENTATION SCREENSHOTS

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- Admin Login Page

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- The administrator is required to enter valid login credentials to access the system.
- Proper validation is implemented to ensure that both fields are not empty and that the entered details match the records stored in the database

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8(B)

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IMPLEMENTATION SCREENSHOTS



- User Login Page

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8(B)

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Register for IdeaForge

Full Name

John Doe

Email Address

john.doe@university.edu

Student ID

S12345678

Send OTP

Cancel



IMPLEMENTATION SCREENSHOTS

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- User Login Page

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- The page consists of input fields including Full Name, Email Address, and Student ID.
- These fields are used to uniquely identify the participant and ensure proper communication.

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8(B)

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IMPLEMENTATION SCREENSHOTS

- Module 1 Screenshot

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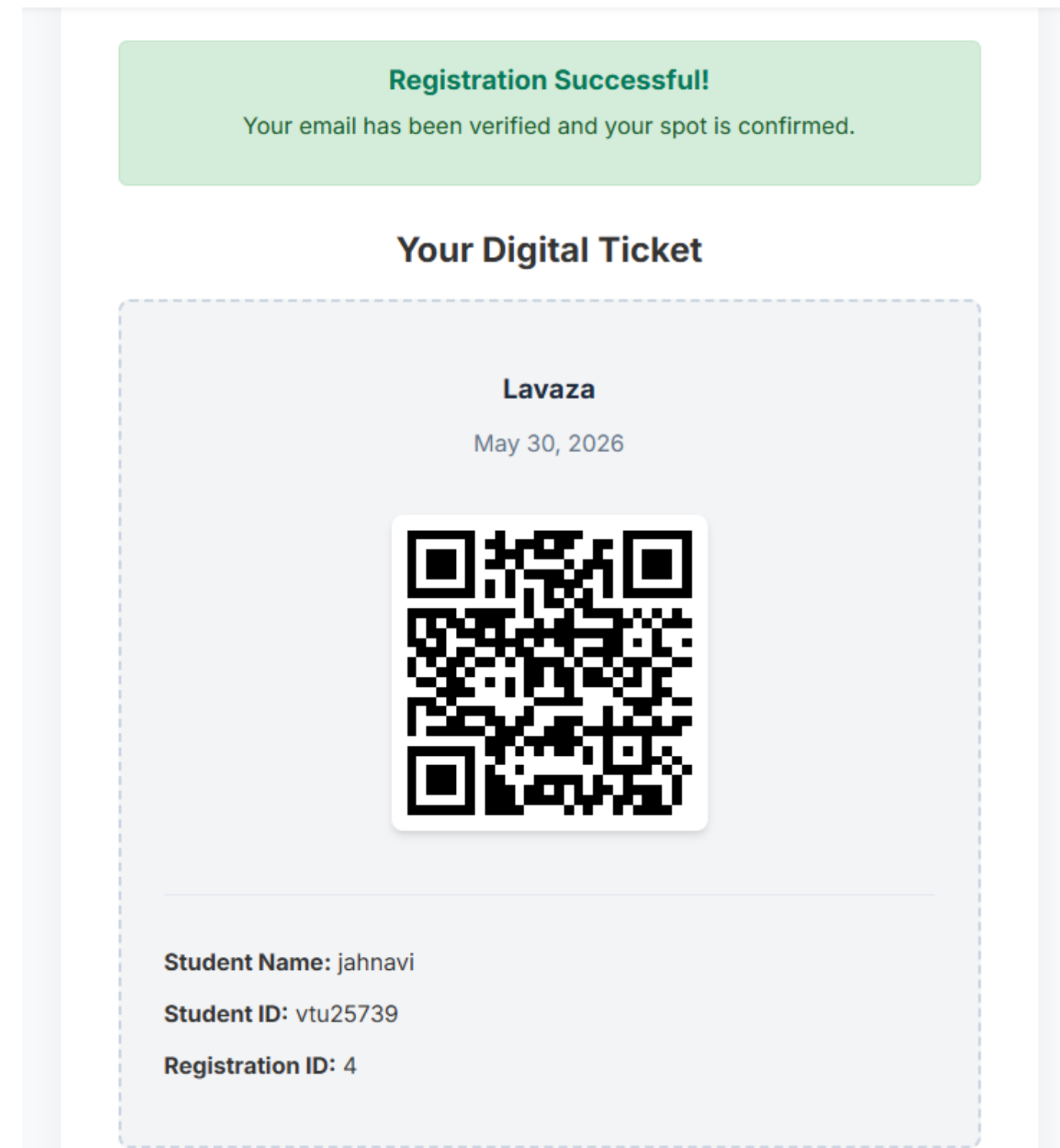
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8(C)

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IMPLEMENTATION SCREENSHOTS

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- Module 1 Explanation

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8(C)

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- This page shows that the event registration is successful.
- It confirms that the user's email has been verified and the seat has been booked. It also displays a digital ticket for the event, including the event name, date, and a QR code.
- The QR code can be used for entry or verification at the event.
- Additionally, the page contains student details such as name, student ID, and registration ID for identification purposes.

IMPLEMENTATION SCREENSHOTS



- Module 2 Screenshot

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8(C)

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Smart Campus Admin

Logout

Create New Event

Event Title

Enter event title

Event Date

dd-mm-yyyy

Event Type

Select Type

Department

e.g. Computer Science

Maximum Capacity

Venue Location

e.g. Main Auditorium, Building B

Description

Enter event details...

Save Event

Cancel



IMPLEMENTATION SCREENSHOTS

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- Module 2 Explanation

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- The Add Event page allows users to create and add new event details in the Smart Campus Event Management System.
- It collects information such as event name, date, location, and status, which are stored in the database for event management.

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8(C)

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IMPLEMENTATION SCREENSHOTS

- Innovative Module Screenshot

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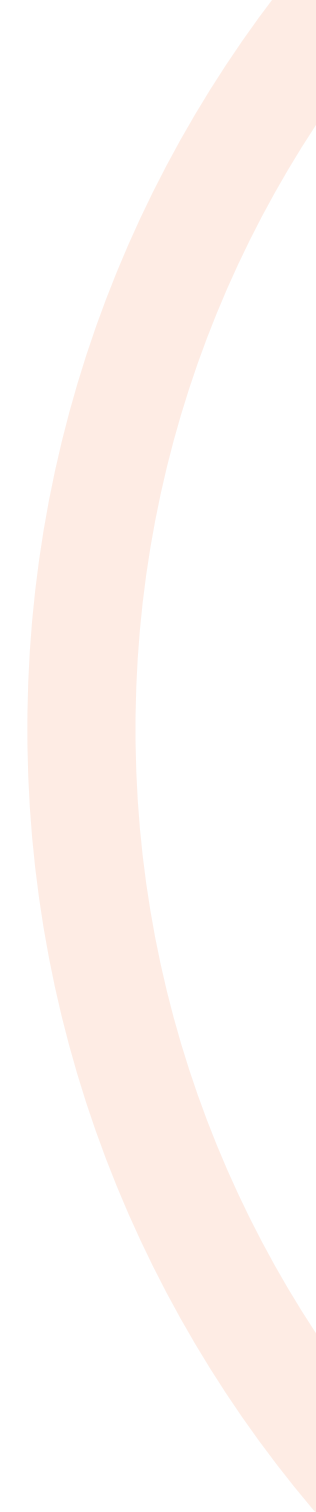
Seminar IT

Tantraz

Date: June 10, 2026 **Capacity:** 0 / 50 **Location:** Vel Murugan Auditorium

About this event

Tantraz





IMPLEMENTATION SCREENSHOTS

- Innovative Module Screenshot

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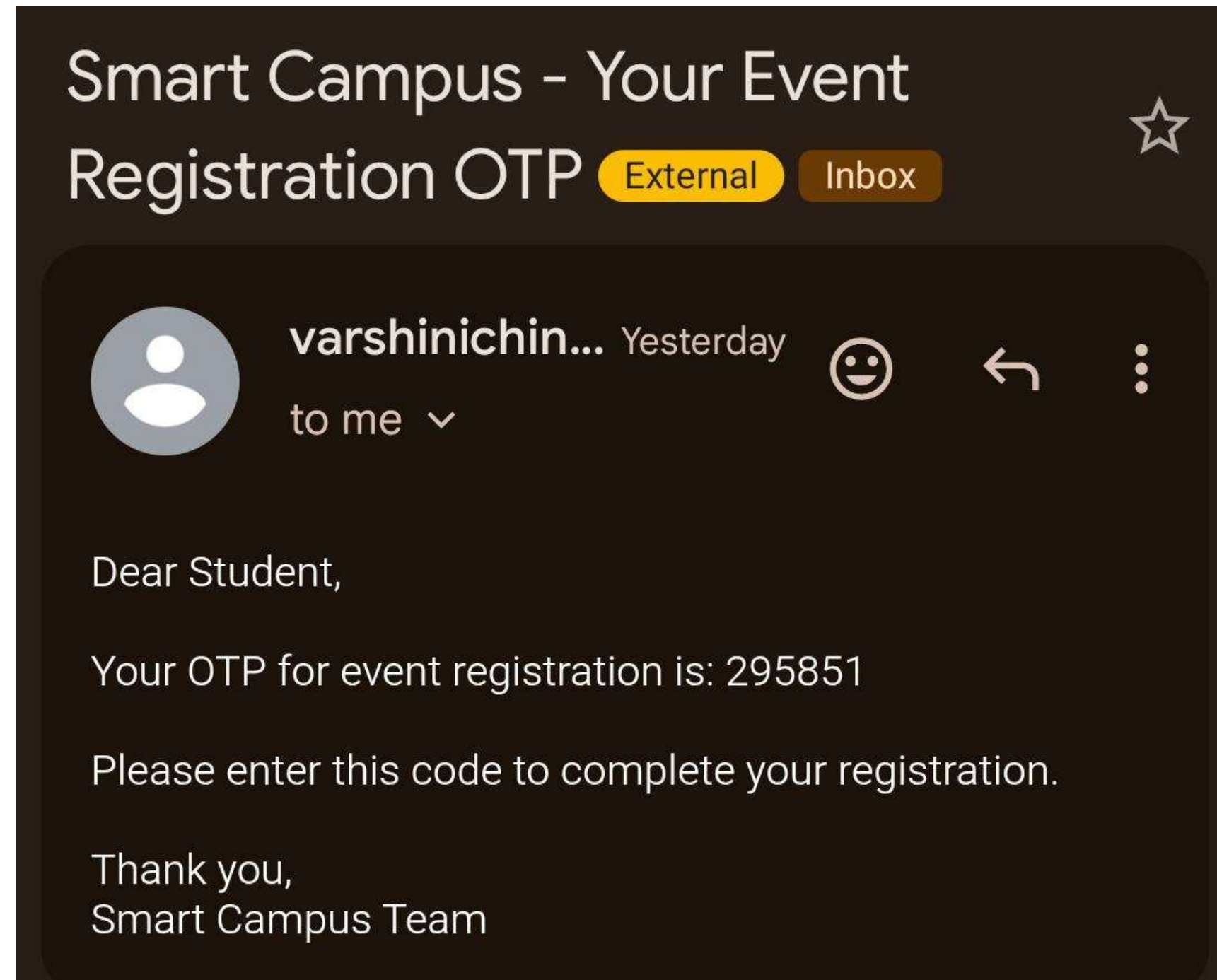
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IMPLEMENTATION SCREENSHOTS

- Innovative Module Screenshot

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8(C)

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Verify Your Email

We've sent a 6-digit OTP to your email address. Please enter it below to complete your registration.

Enter OTP Code

Verify & Register

Cancel Registration



IMPLEMENTATION SCREENSHOTS

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8(C)

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- **Innovative Module** Explanation

- The system incorporates several advanced features to enhance security and user experience.
- An OTP (One-Time Password) verification mechanism is implemented to ensure that only genuine users can register for events, reducing the chances of fake registrations.
- The OTP is securely sent to the user's email, adding an extra layer of authentication.
- Additionally, location-based functionality is integrated to provide users with event venue details and improve accessibility.
- Together, these features make the system more secure, reliable, and user-friendly while offering a smarter event management experience.



IMPLEMENTATION SCREENSHOTS

- Database Screenshot

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8(C)

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The screenshot displays a database management interface. On the left, a 'Navigator' pane shows a tree structure of schemas. The 'event_management' schema is expanded, showing tables like 'event', 'events', 'registration', and 'registrations'. The 'events' table is selected. The main area shows a query window with the SQL statement 'SELECT * FROM events'. Below the query, a 'Result Grid' displays the data. The grid has columns: id, department, description, event_date, event_type, max_capacity, title, and location. The data rows are:

	id	department	description	event_date	event_type	max_capacity	title	location
▶	2	CSE	IdeaForge	2026-05-05	Seminar	28	IdeaForge	CK Naidu Stadium
	3	CSE	Lavaza	2026-05-30	Conference	60	Lavaza	Convocation Hall
	4	IT	Tantraz	2026-06-10	Seminar	50	Tantraz	Vel Murugan Auditorium
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

At the bottom, there are buttons for 'Apply' and 'Revert'.



IMPLEMENTATION SCREENSHOTS

- Database Screenshot

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8(C)

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The screenshot displays a database management interface. On the left, the 'Navigator' pane shows the 'event_management' schema with tables 'event', 'events', 'registration', and 'registrations'. The 'registrations' table is selected. The main area shows 'Query 1' with the SQL statement: `SELECT * FROM registrations`. Below the query, the 'Result Grid' displays the following data:

	id	registration_time	student_email	student_id	student_name	event_id
▶	2	2026-05-04 21:27:34.678473	vtu25643@veltech.edu.in	vtu25643	Hanvitha	2
	3	2026-05-04 21:40:35.164357	varshinichinta123@gmail.com	vtu24829	Varshini	2
	4	2026-05-05 00:36:14.790926	mommineedijahnavisatya@gmail.com	vtu25739	jahnavi	3
	5	2026-05-05 22:24:45.643055	varshinichinta123@gmail.com	vtu24829	Varshini	3
*	NULL	NULL	NULL	NULL	NULL	NULL

The interface also includes a 'Form Editor' button on the right and 'Apply' and 'Revert' buttons at the bottom right.

ADVANTAGES & LIMITATIONS



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Advantages

- Centralized system for managing all campus events
- Reduces manual work and paperwork
- Easy event registration for students
- Improves communication and participation
- Provides real-time data and analytics
- Secure and scalable application
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Limitations

- Requires internet connectivity to access the system
- Initial setup and development cost is high
- Requires technical knowledge for maintenance
- Dependent on system/server availability
- Limited features without further upgrades

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Thank you